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Cheng et al.

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(54) **DISPLAY MODULE AND DISPLAY DEVICE**

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CPC G02F 1/133608; G02F 1/133603; G02F 1/133607
See application file for complete search history.

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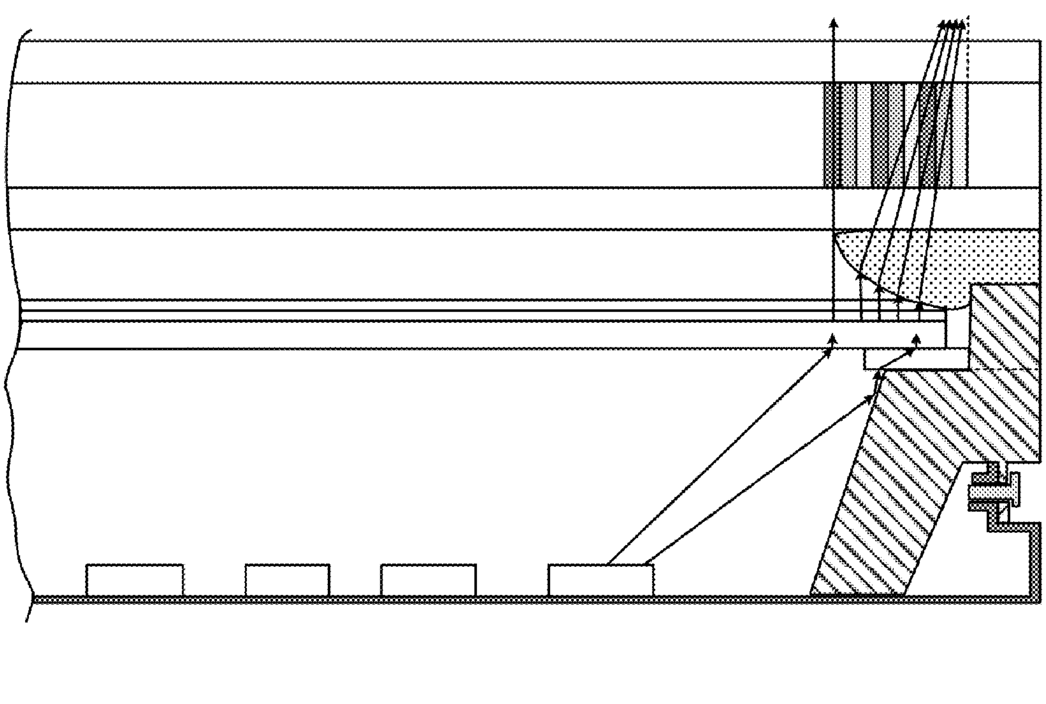
(57) **ABSTRACT**

Provided is a display module. The display module includes: a display panel; an adhesive layer; and a backlight module including a backlight source, a frame body, and an optical film, wherein the frame body includes a bearing stage, a support stage, and a barrier wall, wherein the bearing stage includes a bearing face configured to bear the optical film, the support stage is affixed to the bearing face, and the barrier wall is affixed to a face, facing away from the bearing face, of the support stage; wherein the adhesive layer includes a first adhesive portion and a second adhesive portion.

19 Claims, 9 Drawing Sheets



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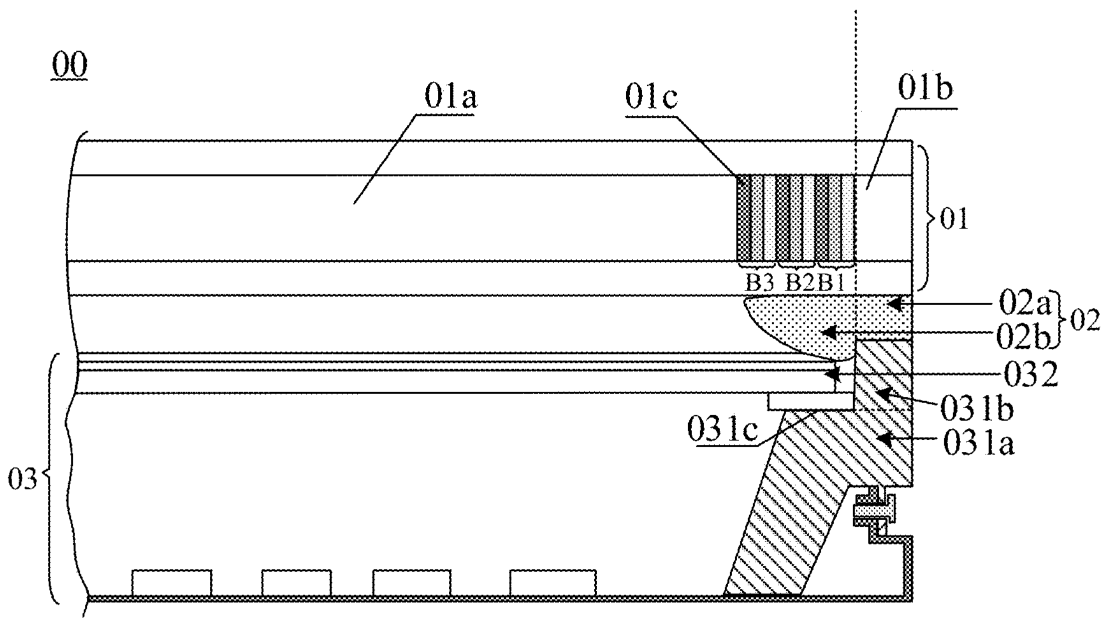

FIG. 1

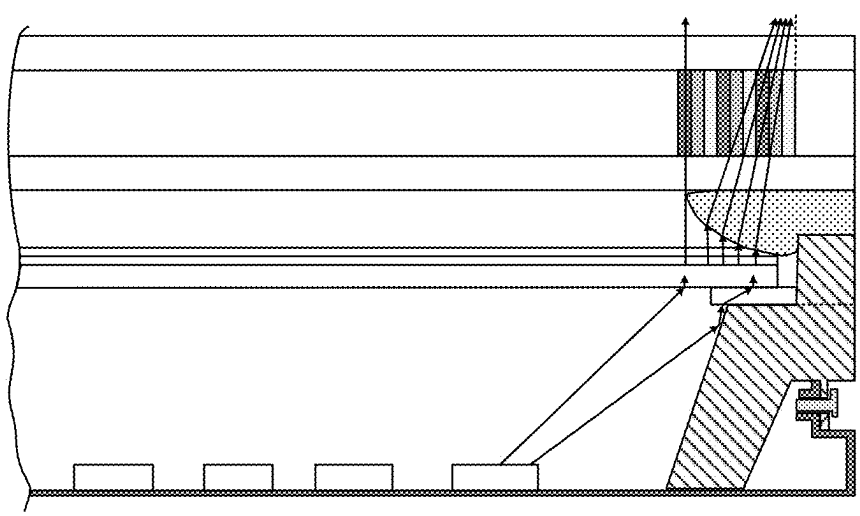

FIG. 2

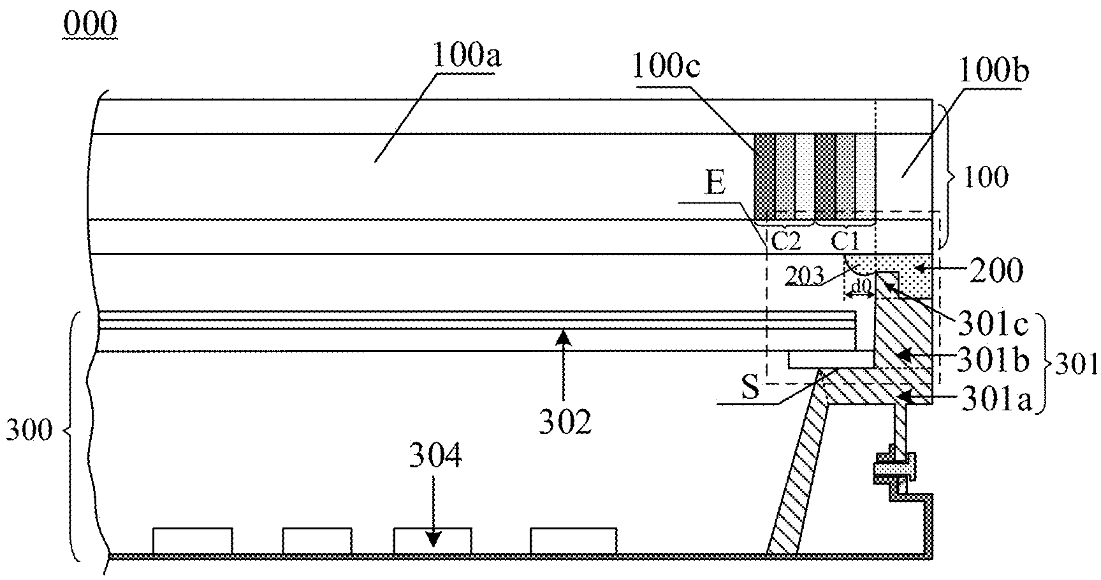
000
100a
100c
100b
100
E
C2
C1
200
203
d0
301c
301b
301
301a
S
302
300
304

FIG. 3

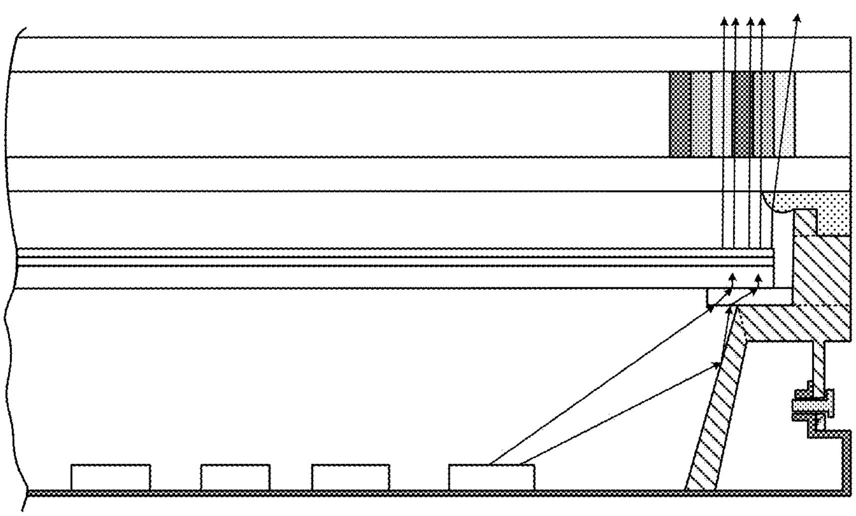

FIG. 4

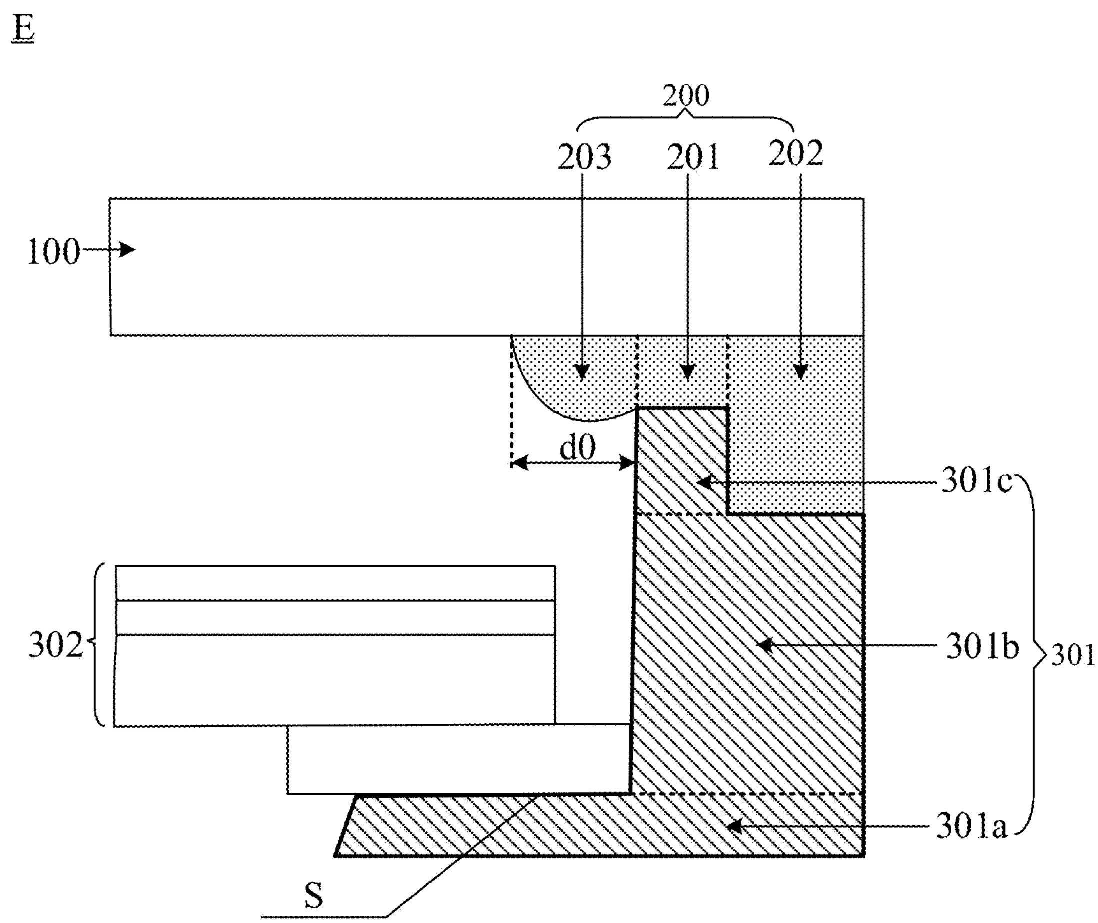

FIG. 5

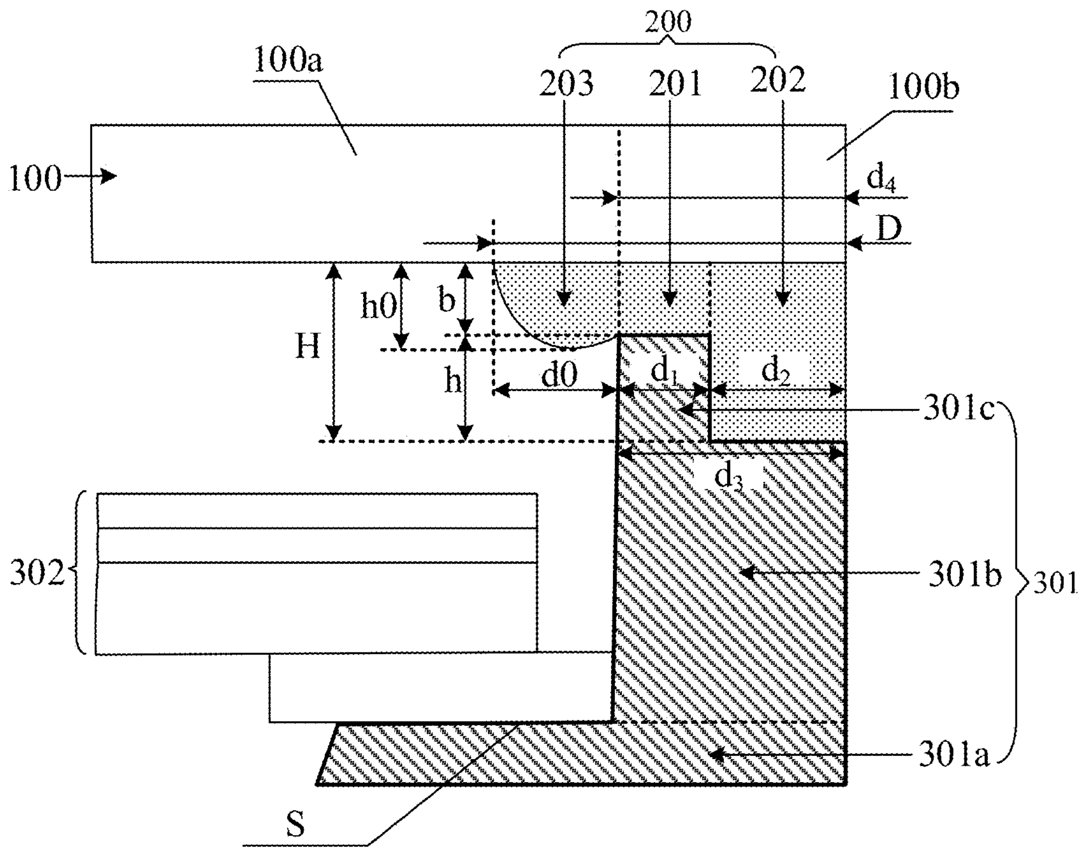
200
100a
203
201
202
100b
100
d4
D
h0
b
H
h
d0
d1
d2
301c
d3
302
301b
301
301a
S

FIG. 6

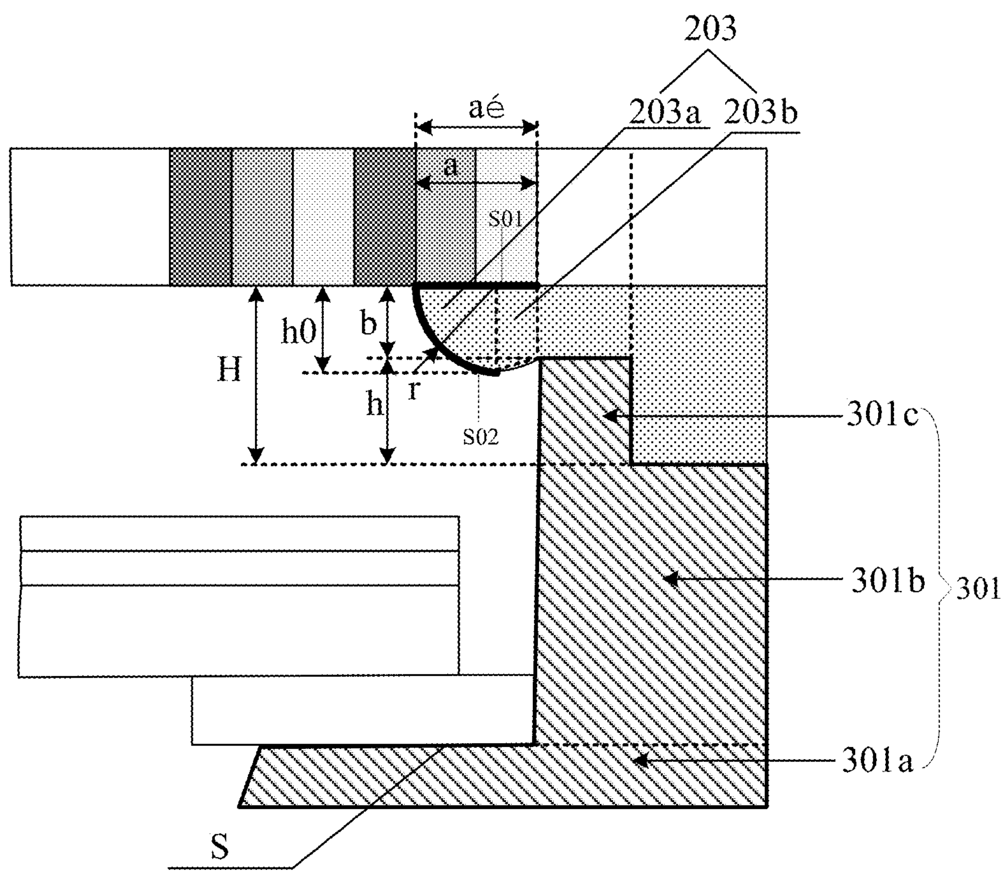

FIG. 7

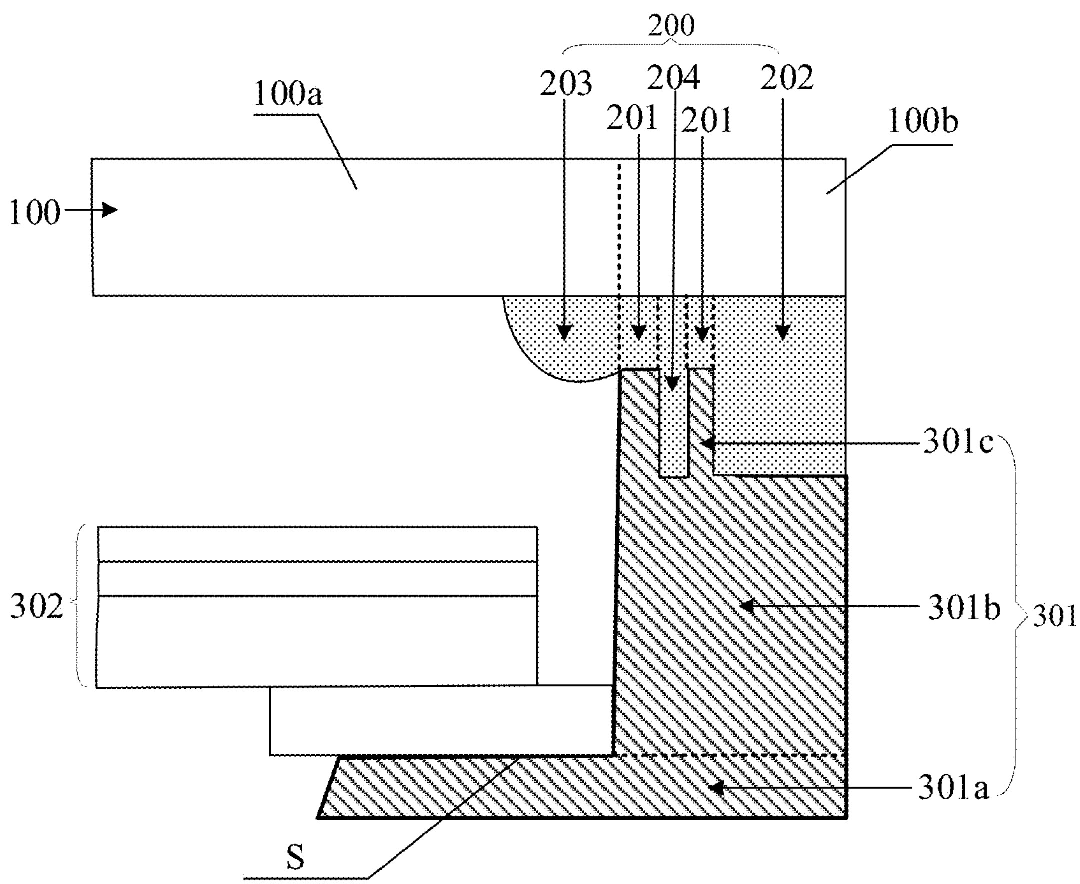

FIG. 8

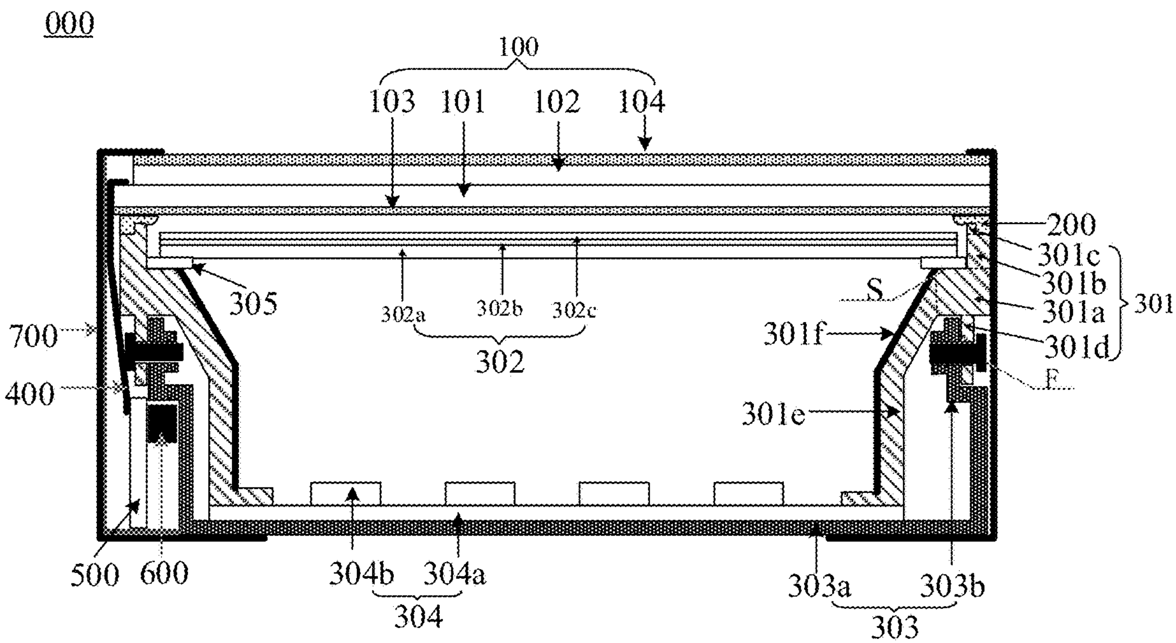
000
100
103
101
102
104
200
301c
301b
301a
301
301d
305
302a
302b
302c
302
S
301f
700
400
E
301e
500
600
304b
304a
304
303a
303b
303

FIG. 9

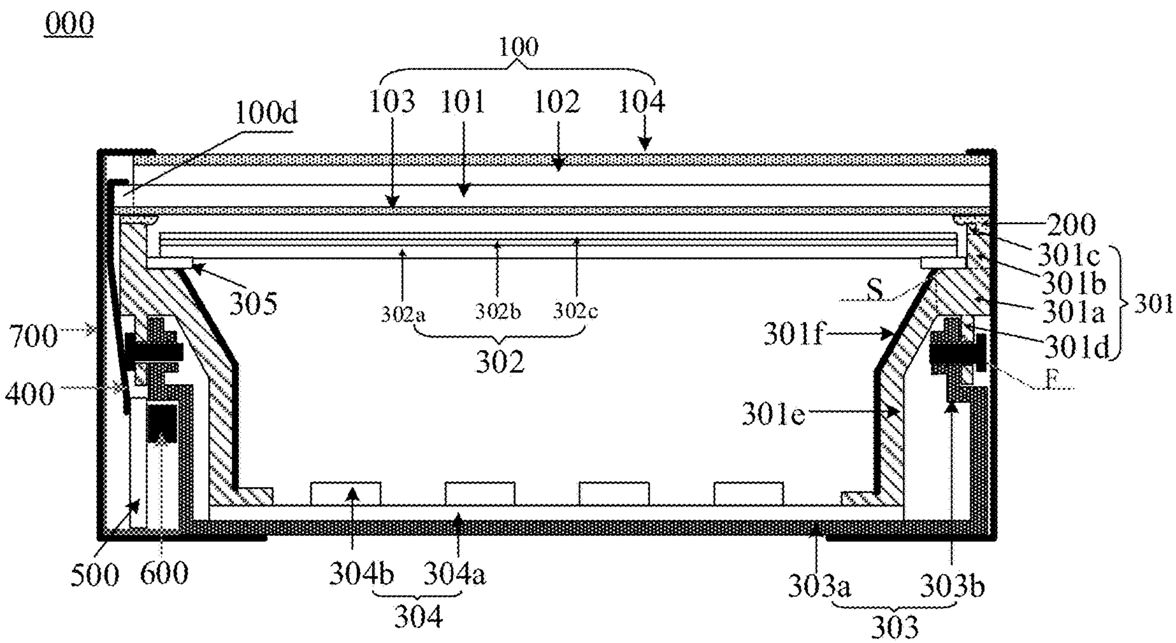
000
100
103
101
102
104
100d
200
301c
301b
301a
301
301d
305
S
302a
302b
302c
302
700
301f
F
400
301e
500
600
304b
304a
304
303a
303b
303

FIG. 10

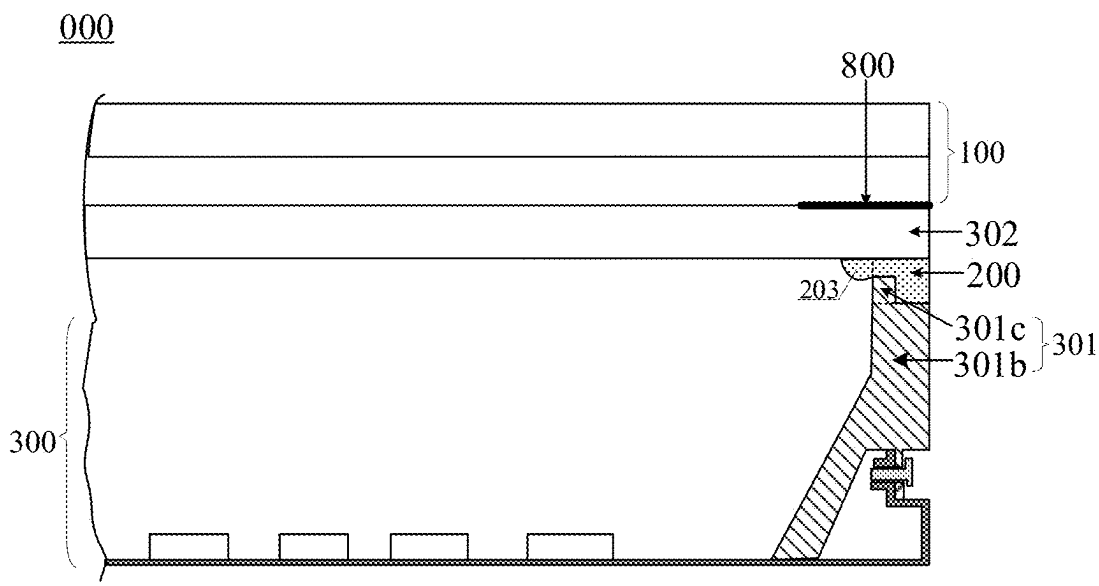
000
800
100
302
200
203
301c
301
301b
300

FIG. 11

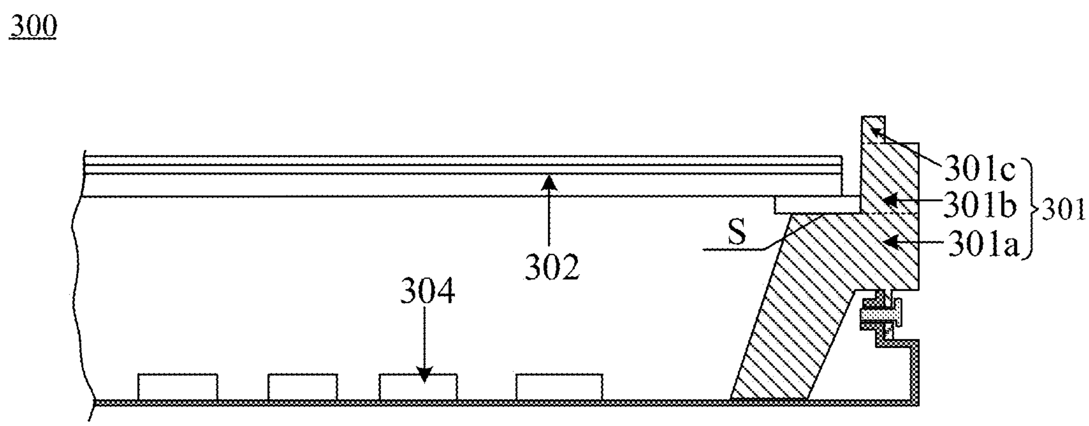
300
301c
301b
301
301a
S
302
304

FIG. 12

DISPLAY MODULE AND DISPLAY DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

This application is a U.S. national stage of international application No. PCT/CN2022/115964, filed on Aug. 30, 2022, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to the field of display technologies, and in particular, relates to a display module and a display device.

BACKGROUND OF THE INVENTION

For enhancement of comfort of users during use of display devices, narrow-frame display devices with a larger screen and a narrow frame have been developed. In some practices, the display device generally includes a backlight module and a liquid crystal display panel. The backlight modules include a direct backlight module and a side-edge backlight module. The narrow frame display device is easily acquired based on the direct backlight module.

SUMMARY OF THE INVENTION

Embodiments of the present disclosure provide a display module and a display device. The technical solutions are as follows.

In some embodiments of the present disclosure, a display module is provided. The display module includes:

a display panel;

an adhesive layer; and

a backlight module including a backlight source, a frame body, and an optical film, wherein

the frame body includes a bearing stage, a support stage, and a barrier wall, wherein the bearing stage includes a bearing face configured to bear the optical film, the support stage is affixed to the bearing face, and the barrier wall is affixed to a face, facing away from the bearing face, of the support stage; wherein

the adhesive layer includes a first adhesive portion and a second adhesive portion, wherein the first adhesive portion is disposed between the barrier wall and the display panel and is bonded to the barrier wall and the display panel, the second adhesive portion is disposed between the support stage and the display panel and is bonded to the support stage and the display panel, and

the second adhesive portion is closer to an edge of the display module than the first adhesive portion is.

In some embodiments, a minimum width d_1 of a cross-section of the barrier wall and a width d_3 of a side, close to the barrier wall, of a cross-section of the support stage meet: $d_1 \leq d_3/2$.

In some embodiments, a face, close to the optical film, of the barrier wall is flush with a face, close to the optical film, of the support stage.

In some embodiments, the minimum width d_1 of the cross-section of the barrier wall and the width d_3 of the side, close to the barrier wall, of the cross-section of the support stage meet: $d_1 \geq d_3/4$.

In some embodiments, in a direction perpendicular to a plane of a light exiting face of the display panel, a minimum height h of the barrier wall and a minimum distance H between the support stage and the display panel meet: $H/4 \leq h \leq 3H/4$.

In some embodiments, in a direction perpendicular to a plane of a light exiting face of the display panel, a minimum distance between the display panel and the support stage is greater than or equal to 0.2 mm; and/or, in the direction perpendicular to the plane of the light exiting face of the display panel, the minimum distance between the display panel and the support stage is less than or equal to 0.5 mm.

In some embodiments, a maximum width of a cross-section of the adhesive layer is greater than or equal to 1 mm.

In some embodiments, the adhesive layer further includes an overflowed adhesive portion on a side, facing away from the second adhesive portion, of the first adhesive portion, wherein the overflowed adhesive portion is bonded to the display panel; and

the display panel includes a plurality of valid sub-pixels arranged in an array, and a maximum width of a cross-section of the overflowed adhesive portion is less than or equal to an overall width of two valid sub-pixels closest to an edge of a display region of the display panel.

In some embodiments, in a direction perpendicular to a plane of a light exiting face of the display panel, a maximum thickness of the overflowed adhesive portion is less than or equal to twice a minimum distance between the display panel and the barrier wall.

In some embodiments, the cross-section of the overflowed adhesive portion includes an adhesive face and an arc-shaped face, wherein the adhesive face is bonded to the display panel, the arc-shaped face is disposed on a side, facing away from the display panel, of the adhesive face, and a maximum distance between the arc-shaped face and the display panel is less than or equal to a maximum distance between the support stage and the display panel.

In some embodiments, an area S1' of the cross-section of the overflowed adhesive portion and an area S2 of a cross-section of the barrier wall meet:

$$\frac{\pi\times\left(\frac{H}{0.625}\right)^2}{4} - S1' \leqslant S2';$$

wherein H represents a minimum distance between the support stage and the display panel.

In some embodiments, relevant dimensions of the barrier wall meet:

$$\begin{cases} S1 = \frac{\pi r^2}{4} + \frac{1}{2}\times(b+r)\times(a-r) \\ r = h0 \\ S2 = d_1 \times h \\ \frac{\pi\times\left(\frac{H}{0.625}\right)^2}{4} - S1 \leqslant S2 \end{cases};$$

wherein H represents a minimum distance between the support stage and the display panel, d_1 represents a minimum width of a cross-section of the barrier wall, h represents a minimum height of the barrier wall in a direction perpendicular to a plane of a light exiting face of the display panel, r represents a radius of the arc-shaped face, h0 represents a maximum thickness of the overflowed adhesive portion, b represents a minimum distance between the barrier wall and the display panel, and a represents the overall width of two valid sub-pixels closest to the edge of the display region.

In some embodiments, relevant dimensions of the barrier wall meet:

$$\begin{cases} S1 = \frac{\pi r^2}{4} + \frac{1}{2} \times (b + r) \times (a' - r) \\ r = h0 \\ S2 = d_1 \times h \\ \frac{\pi \times \left(\frac{H}{0.625}\right)^2}{4} - S1 \leqslant S2 \end{cases};$$

wherein H represents a minimum distance between the support stage and the display panel, d_1 represents a minimum width of a cross-section of the barrier wall, h represents a minimum height of the barrier wall in a direction perpendicular to a plane of a light exiting face of the display panel, r represents a radius of the arc-shaped face, h0 represents a maximum thickness of the overflowed adhesive portion, b represents a minimum distance between the barrier wall and the display panel, and a' represents a distance between an orthogonal projection of an edge, away from the edge of the display region, of a second valid sub-pixel closest to the edge of the display region of the display panel on the display panel and an orthogonal projection of the barrier wall on the display panel.

In some embodiments, a shear resistance strength P of the adhesive layer meets:

$$P \geqslant \frac{m \times g}{12 \times s};$$

wherein m represents a mass of the display panel, g represents gravitational acceleration, and s represents a minimum area of a cross-section of the adhesive layer.

In some embodiments, the adhesive layer is formed by curing an optical adhesive.

In some embodiments, at least two barrier walls are juxtaposed on a face, facing away from the bearing face, of the support stage, a gap is defined between any two adjacent barrier walls, and

the adhesive layer further includes an auxiliary adhesive portion in the gap.

In some embodiments, the display panel further includes a display region and a non-display region on a periphery of the display region, and a width of a side, close to the barrier wall, of a cross-section of the support stage is less than or equal to a width of the non-display region.

In some embodiments, a boundary of the display region is flush with a face, close to the optical film, of the support stage.

In some embodiments, the display module further includes: a plurality of barrier walls extending along an edge of the display module, wherein

the plurality of barrier walls are disposed on a plurality of side faces of the display module, and orthogonal projections of the plurality of barrier walls on the display panel surround the display region of the display panel.

In some embodiments, the non-display region includes a bonding region, wherein the bonding region is disposed on a first side of the display region, and the orthogonal projections of the plurality of barrier walls on the display panel are on other sides of the display region than the first side.

In some embodiments, the backlight module further includes: a light guide structure on the bearing face of the bearing stage, wherein the support stage is closer to the edge of the display module than the light guide structure is, and the optical film is disposed on a side, facing away from the bearing face, of the light guide structure.

In some embodiments, the backlight module further includes: a backplane, wherein the backplane is disposed on a side, facing away from the display panel, of the optical film, the frame body is affixed to the backplane, and the backlight source is disposed between the backplane and the optical film.

In some embodiments, the frame body further includes: a surrounding structure fixedly connected to the bearing stage, wherein the surrounding structure surrounds the backlight source, and a side, close to the backlight source, of the surrounding structure is provided with a reflective layer.

In some embodiments, the optical film includes at least one of: a diffuse plate, a lower prism sheet, and an upper prism sheet.

In some embodiments of the present disclosure, a display module is provided. The display module includes:

a display panel;

an optical film attached to a light entering face of the display panel;

a frame body on a side, facing away from the display panel, of the optical film, wherein the frame body includes: a support stage and a barrier wall, wherein the barrier wall is affixed to a face, facing away from the bearing face, of the support stage;

an adhesive layer, wherein the adhesive layer includes: a first adhesive portion and a second adhesive portion, wherein the first adhesive portion is disposed between the barrier wall and the display panel and is bonded to the barrier wall and the optical film, the second adhesive portion is disposed between the support stage and the display panel and is bonded to the support stage and the optical film, and

the second adhesive portion is closer to an edge of the display module than the first adhesive portion is.

In some embodiments of the present disclosure, a display device is provided. The display device includes a plurality of spliced display modules, wherein the display module includes the above display module.

BRIEF DESCRIPTION OF DRAWINGS

For clearer description of the technical solutions in the embodiments of the present disclosure, the following briefly describes the accompanying drawings required for describing the embodiments. Apparently, the accompanying drawings in the following description show merely some embodiments of the present disclosure, and those of ordinary skill in the art may still derive other drawings from these accompanying drawings without any creative efforts.

FIG. **1** is a schematic structural diagram of a display module;

FIG. **2** is a light path of light emitted by a backlight module in the display module shown in FIG. **1**;

FIG. **3** is a schematic structural diagram of a display module according to some embodiments of the present disclosure;

FIG. **4** is a light path of light emitted by a backlight module in the display module shown in FIG. **3**;

FIG. **5** is a partially enlarged diagram of the display module shown in FIG. **3** at a position of E;

FIG. **6** is a partially enlarged diagram of another display module according to some embodiments of the present disclosure;

FIG. **7** is a partially enlarged diagram of another display module according to some embodiments of the present disclosure;

FIG. **8** is a partially enlarged diagram of another display module according to some embodiments of the present disclosure;

FIG. **9** is a schematic structural diagram of another display module according to some embodiments of the present disclosure;

FIG. **10** is a schematic structural diagram of another display module according to some embodiments of the present disclosure;

FIG. **11** is a schematic structural diagram of another display module according to some embodiments of the present disclosure; and

FIG. **12** is a schematic structural diagram of a backlight module according to some embodiments of the present disclosure.

DETAILED DESCRIPTION OF THE INVENTION

To make the objectives, technical solutions, and advantages of the present disclosure clearer, the embodiments of the present disclosure are further described in detail hereinafter with reference to the accompanying drawings.

The conventional direct backlight module generally includes an optical film, a light plate, a frame body, and a backplane. The optical film and the light plate are generally laminated, the frame body is configured to wrap the optical film and the light plate, and the backplane is configured to support the light plate. In assembling the backlight module and the display panel, an adhesive layer requires to be disposed between the display panel and the frame body in the backlight module, and the adhesive layer is separately bonded to the display panel and the frame body.

However, a frame of the current display device is becoming increasingly narrow, and a distance between an outer boundary of a display region of the display device and an inner boundary of the frame body in the backlight module is becoming increasingly less. In this case, in bonding the display panel to the frame body by the adhesive layer, the adhesive layer between the display panel and the frame body is prone to adhesive overflow, and the overflowed adhesive is distributed in the display region, such that a display effect of the display device is poor.

Referring to FIG. **1**, FIG. **1** is a schematic structural diagram of a display module. The display module **00** includes a display panel **01**, an adhesive layer **02**, and a backlight module **03**.

The display panel **01** includes a display region Ola and a non-display region **01***b* on a periphery of the display region Ola. The display region Ola of the display panel **01** is coincided with a display region of the display module. The display region Ola of the display panel **01** includes a plurality of sub-pixel region **01***c* arranged in an array. One sub-pixel is disposed in each sub-pixel region **01***c*. Each three adjacent sub-pixels in the display panel **01** are configured to compose one pixel. For example, three pixels on an edge portion of the display panel **01** in FIG. **1** are a pixel B1, a pixel B2, and a pixel B3, and the pixel B1 is closest to the non-display region **01***b*. Pixels in the row of the pixel B1 are a first row of pixels close to the non-display region **01***b*, pixels in the row of the pixel B2 are a second row of pixels close to the non-display region **01***b*, and pixels in the row of the pixel B3 are a third row of pixels close to the non-display region **01***b*.

The backlight module **03** includes a frame body **031** and an optical film **032**. The frame body **03** includes a bearing stage **031***a* and a support stage **031***b*. The bearing stage **031***a* is provided with a bearing face **031***c* for bearing the optical film **032**, and the support stage **031***b* is affixed to the bearing face **031***c* of the bearing stage **031***a*.

In assembling the backlight module **03** and the display panel **01**, an adhesive layer **02** requires to be disposed between the display panel **01** and the support stage **031***b* in the frame body **031** of the backlight module **03**. As such, the display panel **01** is bonded to the support stage **031***b* in the frame body **031** by the adhesive layer **02**, such that connection between the display panel **01** and the backlight module **03** is achieved.

However, a frame of the current display module is increasingly narrow, a boundary of the display region of the display module is flush with a face, close to the optical film **032**, of the support stage **031***b* in the frame body **031**, and a face, facing away from the optical film **032**, of the support stage **031***b* does not exceed the display panel **01**. Thus, a width of the support stage **031***b* is less, and the adhesive layer **02** between the display panel **01** and the support stage **031***b* is prone to adhesive overflow in bonding the display panel **01** and the support stage **031***b* by the adhesive layer **02**. For example, the adhesive layer **02** includes an adhesive portion **02***a* between the display panel **01** and the support stage **031***b* and an overflowed adhesive portion **02***b* on a side of the adhesive portion **02***a*. The overflowed adhesive portion **02***b* in the adhesive layer **02** is disposed in the display region of the display module, and a width of the overflowed adhesive portion **02***b* is generally great. For example, the width of the overflowed adhesive portion **02***b* is generally greater than a width of two pixels (that is, six sub-pixel regions **01***c*). Thus, in bonding the display panel **01** and the support stage **031***b* by the adhesive layer **02**, the overflowed adhesive of the overflowed adhesive portion **02***b* in the adhesive layer **02** stops in the case that the overflowed adhesive portion **02***b* in the adhesive layer **02** reaches the third row of pixels close to the non-display region **01***b*.

In the case that the width of the overflowed adhesive portion **02***b* in the adhesive layer **02** is great, the overflowed adhesive portion **02***b* in the adhesive layer **02** affects the display effect of the display panel **01**. For example, referring to FIG. **2**, FIG. **2** is a light path of light emitted by a backlight module in the display module shown in FIG. **1**. A face, facing away from the display panel **01**, of the overflowed adhesive portion **02***b* in the adhesive layer **02** is a curved face, light emitted from the backlight module **03** to the second row of pixels close to the non-display region **01***b* is refracted upon passing thorough the curved face, and most refracted light is exited from the first row of pixels close to the non-display region **01***b*. As such, light emitted from the backlight module **03** and exited from the second row of pixels close to the non-display region **01***b* is less, such that the poor phenomenon of dark lines is caused in the edge portion of the screen displayed by the display panel **01**. As such, the display effect of the display panel **01** is poor.

Referring to FIG. **3**, FIG. **3** is a schematic structural diagram of a display module according to some embodiments of the present disclosure. The display module **000** includes: a display panel **100**, an adhesive layer **200**, and a backlight module **300**.

The backlight module **300** in the display module **000** includes: a backlight source **304**, a frame body **301**, and an optical film **302**. The frame body **301** surrounds the backlight source **304** and the optical film **302**, and the optical film **302** is connected to the frame body **301**. For example, the optical film **302** homogenizes the backlight.

The adhesive layer **200** in the display module **000** is bonded to the frame body **301** and the display panel **100**. The optical film **302** in the backlight module **300** is closer to the display panel **100** than the backlight source **304** is. The adhesive layer **200** is formed by curing the optical adhesive. For example, the optical adhesive is viscous. Thus, in assembling the frame body **301** and the display panel **100** by the adhesive layer **200**, the overflowed adhesive occurs in the adhesive layer **200**. For example, the adhesive layer **200** includes adhesive portions (that is, a first adhesive portion **201** and a second adhesive portion **202**) between the display panel **100** and the frame body **301** and an overflowed adhesive portion **203** on a side of the adhesive portion. The overflowed adhesive portion **203** of the adhesive layer **200** is disposed in the display region of the display module.

The display panel **100** in the display module **000** includes a display region **100***a* and a non-display region **100***b* on a periphery of the display region **100***a*. The display region **100***a* of the display panel **100** is coincided with the display region of the display module. The display panel **100** includes: a plurality of valid sub-pixels **100***c* arranged in an array. The plurality of valid sub-pixels **100***c* are arranged in the display region **100***a*. Each three adjacent valid sub-pixels compose one pixel. For example, three valid sub-pixels in each pixel are a red sub-pixel, a green sub-pixel, and a blue sub-pixel. Both the pixel C1 and the pixel C2 in the edge portion of the display panel **100** in FIG. **3** include three sub-pixels, and the pixel C1 is closer to the non-display region **100***b*. Pixels in the row of the pixel C1 are a first row of pixels close to the non-display region **100***b*, and pixels in the row of the pixel C2 are a second row of pixels close to the non-display region **100***b*. The first row of pixels is a row of pixels in the display region **100***a* closest to an edge of the display region **100**, and the second row of pixels is a row of pixels in the display region **100***a* closest to an edge of the display region **100** after the first row of pixels.

It should be noted that in bonding the display panel **100** and the backlight module **300** by the adhesive layer **200**, for amounts of the overflowed adhesive at different positions of the adhesive layer **200**, other than less positions (for example, a starting position of dispensing of the dispensing process for forming the adhesive layer **200**), amounts of the overflowed adhesive at most positions are the same. Thus, in the overflowed adhesive portion **203** of the adhesive layer **200**, shapes and sizes of cross-sections of most positions other than these less positions are the same. The cross-section of the overflowed adhesive portion **203** in the following embodiments refers to the cross-sections at these most positions.

In some embodiments, on at least one side of the display module **000**, a distance between an orthogonal projection of a boundary of a side, away from an edge of the display module **000**, of the adhesive layer **200** on the display region **100***a* and the edge of the display region **100***a* is less than or equal to a distance between an edge, away from the display region **100***a*, of the first row of pixels closest to the edge of the display region **100***a* and the edge of the display region **100***a*.

In some embodiments, on at least one side of the display module **000**, the distance between the orthogonal projection of the boundary of the side, away from the edge of the display module **000**, of the adhesive layer **200** on the display region **100***a* and the edge of the display region **100***a* is less than or equal to a distance between an edge, away from the display region **100***a*, of the second valid sub-pixel **100***c* closest to the edge of the display region **100***a* and the edge of the display region **100***a*. In some embodiments, the distance between the orthogonal projection of the boundary of the side, away from the edge of the display module **000**, of the adhesive layer **200** on the display region **100***a* and the edge of the display region **100***a* is greater than a distance between an edge, away from the display region **100***a*, of the first valid sub-pixel **100***c* closest to the edge of the display region **100***a* and the edge of the display region **100***a*. As such, the bonding strength of the structure is ensured on the premise that the optical effect is not affected. In some embodiments, the valid sub-pixel **100***c* is in a strip shape, and at least one side of the display module **000** includes a side on which an extension direction of the adhesive layer **200** is the same as an extension direction of the strip sub-pixel. For example, the valid sub-pixel **100***c* is in a rectangular shape including long sides and short sides, and at least one side of the display module **000** includes a side on which the extension direction of the adhesive layer **200** is the same as an extension direction of the long side of the rectangle. In some embodiments, at least one side of the display module **000** includes two opposite sides of the display module **000**.

In some embodiments, on each side of the display module **000**, distances between the orthogonal projections of the edges, away from the display module **000**, of the adhesive layer **200** on the display region **100***a* and the edge of the display region **100***a* are equal.

In some embodiments, a maximum width d0 of a cross-section of the overflowed adhesive portion **203** is less than or equal to an overall width of two valid sub-pixels **100***c* closest to the edge of the display region **100***a*. It should be noted that widths of cross-sections of the overflowed adhesive portion **203** at different heights are different, and thus there is a height at which an area of the cross-section of the overflowed adhesive portion **203** is maximum, and the width of the cross-section at the position is the maximum width of the cross-section of the overflowed adhesive portion **203**. For example, in FIG. **3**, the width of the cross-section of the overflowed adhesive portion **203** at a position closest to the display panel **100** is maximum. The overall width of two valid sub-pixels **100***c* is an overall distance in the width direction of two continuous sub-pixels **100***c*. For example, a shape of the valid sub-pixel **100***c* is a rectangle including long sides and short sides, and a width direction is an extension direction of the short side of the valid sub-pixel **100***c*.

A cross-section of a structure in the embodiments of the present disclosure is a cross-section perpendicular to a plane of a light-exiting face of the display panel **100** and perpendicular to an extension direction of the structure. For example, the cross-section of the overflowed adhesive portion **203** is a cross-section perpendicular to the plane of the light-exiting face of the display panel **100** and perpendicular to the extension direction of the overflowed adhesive portion **203**.

It should be further noted that the display panel **100** is generally provided with a black matrix including a plurality of light holes, and the plurality of light holes in the black matrix are in one-to-one correspondence to the plurality of valid sub-pixels **100***c* in the display panel **100**. With such design, a region of the valid sub-pixel **100***c* includes a region of the corresponding light hole and a region of a light shielding portion surrounding the light hole. The light shielding portion surrounding the light hole is a portion of the black matrix, and a width of the light shielding portion surrounding the light hole is equal to half a width of a black matrix between two adjacent light holes. Thus, the width of the valid sub-pixel **100***c* is equal to a sum of a width of the region of the corresponding light hole, a width of the light shielding portion on one side of the light hole, and a width of the light shielding portion on the other side of the light hole. It should be noted that a width of the light shielding portion on a side, close to the edge of the display region **100***a*, of the valid sub-pixel **100***c* closest to the edge of the display region **100***a* is equal to a width of the light shielding portion on a side, away from the edge of the display region **100***a*, of the valid sub-pixel **100***c* closest to the edge of the display region **100***a*.

In some embodiments, widths of the valid sub-pixels **100***c* are the same, and an overall width of two valid sub-pixels **100***c* closest to the edge of the display region **100***a* is equal to twice the width of one valid sub-pixel **100***c*.

In some embodiments, a boundary of a valid sub-pixel **100***c* closes to the edge of the display region **100***a* is coincided with the edge of the display region **100***a*.

In the embodiments of the present disclosure, in bonding the display panel **100** and the frame body **301** by the adhesive layer **200**, the overflowed adhesive of the overflowed adhesive portion **200***c* in the adhesive layer **200** stops in the case that the overflowed adhesive portion **203** in the adhesive layer **200** reaches the first row of pixels close to the non-display region **01***b*. As such, even if the overflowed adhesive occurs in the adhesive layer **200** in bonding the display panel **100** and the frame body **301** by the adhesive layer **200** in the embodiments of the present disclosure, the maximum width d0 of the cross-section of the overflowed adhesive portion **203** of the adhesive layer **200** is less, and thus the overflowed adhesive portion **203** does not affect the normal display of the display panel **100**.

Referring to FIG. **4**, FIG. **4** is a light path of light emitted by a backlight module in the display module shown in FIG. **3**. In the case that the maximum width d0 of the cross-section of the overflowed adhesive portion **203** of the adhesive layer **200** is less, light emitted from the backlight module **300** to the second row of pixels close to the non-display region **01***b* is not disturbed by the overflowed adhesive portion **203**, and is normally exited from the second row of pixels close to the non-display region **01***b*. In addition, although the light emitted from the backlight module **300** to the first row of pixels close to the non-display region **01***b* is refracted by the overflowed adhesive portion **203**, the refraction of the light by the overflowed adhesive portion **203** is poor as the maximum width d0 of the cross-section of the overflowed adhesive portion **203** is less. Thus, the light emitted to the first row of pixels close to the non-display region **01***b* is also normally exited from the row of pixels. As such, the possibility of poor phenomenon of dark lines in an edge portion of a screen displayed by the display panel **100** is efficiently reduced, and a display effect of the display panel **100** is efficiently improved.

In some embodiments of the present disclosure, referring to FIG. **3** and FIG. **5**, FIG. **5** is a partially enlarged diagram of the display module shown in FIG. **3** at a position of E. The frame body **301** in the backlight module **300** includes a bearing stage **301***a*, a support stage **301***b*, and a barrier wall **301***c*.

The bearing stage **301***a* in the frame body **301** includes a bearing face S for bearing the optical film **302**, the support stage **301***b* in the frame body **301** is affixed to the bearing face S, and the barrier wall **301***c* in the frame body **301** is affixed to a face, facing away from the bearing face S, of the support stage **301***b*.

The adhesive layer **200** includes a first adhesive portion **201** and a second adhesive portion **202**. The first adhesive portion **201** is disposed between the barrier wall **301***c* in the frame body **301** and the display panel **100** and is bonded to the barrier wall **301***c* and the display panel **100**, and the second adhesive portion **202** is disposed between the support stage **301***b* in the frame body **301** and the display panel **100** and is bonded to the support stage **301***b* and the display panel **100**. The overflowed adhesive portion **203** in the adhesive layer **200** is disposed on a side, facing away from the second adhesive portion **202**, of the first adhesive portion **201**, and is bonded to the display panel **100**.

The second adhesive portion **202** is closer to the edge of the display module **000** than the first adhesive portion **201** is. It should be noted that the edge of the display module **000** is a side edge of the display module **000**. For example, in the case that the display module **000** is a rectangular display module, the display module **000** includes four side edges, and a shape enclosed by orthogonal projections of the four side edges on a plane of a light exiting face of the display panel **000** is a rectangle. Thus, the barrier wall **301***c* is disposed on a side, away from the edge of the display module **000**, of the support stage **301***b*.

In the embodiments of the present disclosure, by disposing the barrier wall **301***c* on the side, facing away from the bearing face S, of the support stage **301***b* in the frame body **301**, an amount of overflowed adhesive from the adhesive layer **200** to the display region of the display module **000** is reduced in bonding the display panel **100** and the frame body **301** by the adhesive layer **200**, such that the maximum width d0 of the cross-section of the overflowed adhesive portion **203** subsequently formed in the adhesive layer **200** is less.

In summary, the display module in the embodiments of the present disclosure includes a display panel, an adhesive layer, and a backlight module. In a frame body of the backlight module, a side, facing away form a bearing face of a bearing stage, of a support stage is disposed with a barrier wall. In bonding the display panel and the frame body by the adhesive layer, an amount of overflowed adhesive from the adhesive layer to a display region of the display module is reduced by the barrier wall, such that a maximum width of a cross-section of an overflowed adhesive portion subsequently formed in the adhesive layer is less. For example, the maximum width of the cross-section of the overflowed adhesive portion in the adhesive layer is less than or equal to an overall width of two valid sub-pixels closest to an edge of the display region of the display panel. Thus, light irradiated to each pixel in light emitted from the backlight module is normally emitted from corresponding pixel. As such, a possibility of poor phenomenon of dark lines in an edge portion of a screen displayed by the display panel is efficiently reduced, and a display effect of the display panel is efficiently improved.

In the embodiments of the present disclosure, referring to FIG. **6**, FIG. **6** is a partially enlarged diagram of another display module according to some embodiments of the present disclosure. A ratio of a minimum width d_1 of a cross-section of the barrier wall **301***c* in the frame body **301** to a minimum distance d2 between a face, close to the edge of the display module **000**, of the barrier wall **301***c* and a face, close to the edge of the display module **000**, of the support stage **301***b* is less than or equal to 1. That is, the minimum width d_1 of the cross-section of the barrier wall

301c and a width d_3 of a side, close to the barrier wall **301**c, of a cross-section of the support stage **301**b meet: $d_1 \leq d_3/2$.

In some embodiments, in the case that both a face, away from the edge of the display module **000**, of the barrier wall **301**c in the frame body **301** and a face, away from the edge of the display module **000**, of the support stage **301**b in the frame body **301** are a plane, the face, away from the edge of the display module **000**, of the barrier wall **301**c in the frame body **301** is flush with the face, away from the edge of the display module **000**, of the support stage **301**b in the frame body **301**. In this case, the minimum width d_1 of the cross-section of the barrier wall **301**c and the width d_3 of the side, close to the barrier wall **301**c, of the cross-section of the support stage **301**b meet: d_1, $d_3/4$.

As such, the amount of overflowed adhesive from the adhesive layer **200** to the display region of the display module **000** is efficiently reduced by the barrier wall **301**c, such that the width of the cross-section of the overflowed adhesive portion **203** subsequently formed by curing in the adhesive layer in a direction parallel to the optical film **302** is ensured to be less.

In the present disclosure, in a direction perpendicular to a plane of a light exiting face of the display panel **100**, a minimum height h of the barrier wall **301**c in the frame body **301** is less than a minimum distance H between the display panel **100** and the support stage **301**b in the frame body **301**. In some embodiments, in the direction perpendicular to the plane of the light exiting face of the display panel **100**, the minimum height h of the barrier wall **301**c and the minimum distance H between the support stage **301**b and the display panel **100** meet: $H/4 \leq h \leq 3H/4$. As such, the first adhesive portion **201** in the adhesive layer **200** is disposed between the barrier wall **301**c and the display panel **100**, and the width of the cross-section of the first adhesive portion **201** in the adhesive layer **200** is equal to the minimum width d_1 of the cross-section, closest to the display panel **100**, of the barrier wall **301**c. Furthermore, as the second adhesive portion **202** in the adhesive layer **200** is disposed between the support stage **301**b and the display panel **100**, and the width of the cross-section of the second adhesive portion **202** is equal to the minimum distance d2 between the face, close to the edge of the display module **000**, of the barrier wall **301**c and the face, close to the edge of the display module **000**, of the support stage **301**b, the display panel **100** is bonded to the frame body **301** by the first adhesive portion **201** and the second adhesive portion **202**, such that the display panel **100** is fixedly bonded to the frame body **301**, and the assembly strength of the display panel **100** and the backlight module **300** upon assembling is further great.

In the embodiments of the present disclosure, in the direction perpendicular to the plane of the light exiting face of the display panel **100**, the greater the minimum distance H between the support stage **301**b and the display panel **100**, the thicker the adhesive layer **200** required to be coated. In this case, in bonding the display panel **100** and the frame body **301** by the adhesive layer **200**, the amount of overflowed adhesive from the adhesive layer **200** to the display region of the display module **000** is greater, and thus the maximum width d0 of the cross-section of the overflowed adhesive portion **203** subsequently formed in the adhesive layer is greater. In the direction perpendicular to the plane of the light exiting face of the display panel **100**, the less the minimum distance H between the support stage **301**b and the display panel **100**, the thinner the adhesive layer **200** required to be coated. In this case, in bonding the display panel **100** and the frame body **301** by the adhesive layer **200**, although the amount of overflowed adhesive from the adhesive layer **200** to the display region of the display module **000** is less, the adhesive force of the cured adhesive layer is not enough to affixed the display panel **100** to the frame body **301**, such that the assembly strength of the display panel **100** and the backlight module **300** is less.

Thus, the minimum distance H between the support stage **301**b and the display panel **100** is greater than or equal to 0.2 mm in the direction perpendicular to the plane of the light exiting face of the display panel, and/or, the minimum distance H between the support stage **301**b and the display panel **100** is less than or equal to 0.5 mm in the direction perpendicular to the plane of the light exiting face of the display panel. That is, in the direction perpendicular to the plane of the light exiting face of the display panel, the minimum distance H between the support stage **301**b and the display panel **100** ranges from 0.2 mm to 0.5 mm. As such, the thickness of the adhesive layer **200** required to be coated is not great and not less, such that the display panel **100** is fixedly bonded to the support stage **301**b by the second adhesive portion **202** in the adhesive layer **200** on the premise that the maximum width of the cross-section of the overflowed adhesive portion **203** in the adhesive layer **200** is less, and the display panel is fixedly bonded to the frame body **301**.

It should be noted that a maximum width (or a minimum height) of a cross-section of a structure in the above embodiments includes a width (or a height) in a case that the shape of the cross-section of the structure is irregular. It should be understood that in the case that the cross-section of the structure is regular, widths of the cross-section of the structure at positions are equal (or heights at position are equal), and the maximum width (or the minimum height) is the width (or the height) of the cross-section. For example, in the case that the cross-section of the barrier wall **301**c is a rectangle, widths d_1 of the cross-section of the barrier wall **301**c at different positions are equal, and heights h of the cross-section of the barrier wall **301**c at different positions are also equal. Similarly, the minimum distance between two structures includes a distance in a case that surfaces of the structures is not planar. It should be understood that in the case that the distance between the structures are planar, distances between the two structures at different positions are equal, and the minimum distance is a distance between the two structures. For example, in the case that the face, close to the display panel, of the support stage **301**b is a plane, and a face, close to the frame body **301**, of the display panel **100** is a plane, distances H between the support stage **301**b and the display panel **100** at different positions are equal.

In the embodiments of the present disclosure, a maximum width D of a cross-section of the adhesive layer **200** is greater than or equal to 1 mm. For example, in the case that the cross-section of the barrier wall **301**c is a rectangle, and the face, close to the display panel, of the support stage **301**b is parallel to the plane of the light exiting face of the display panel **100**, the maximum width D of the cross-section of the adhesive layer **200** is equal to a sum of a width of the first adhesive portion **201**, a width of the second adhesive portion **202** (that is, a distance between the face, close to the edge of the display module **000**, of the barrier wall **301**c and the face, close to the edge of the display module **000**, of the support stage **301**b), and the maximum width of the cross-section of the overflowed adhesive portion **203**. As such, the bonding affixation between the display panel **100** and the frame body **301** is further improved. It should be noted that the overflowed adhesive portion **203** in the adhesive layer can also improve the bonding affixation between the display panel **100** and the frame body **301**.

In the present disclosure, in the direction parallel to the optical film **302**, the maximum width d0 of the cross-section of the overflowed adhesive portion **203** in the adhesive layer **200** is less than or equal to the overall width of two valid sub-pixels **100***c* closest to the edge of the display region **100***a*. Thus, in bonding the display panel **100** and the frame body **301** by the adhesive layer **200**, the overflowed adhesive stops in the case that the amount of overflowed adhesive from the adhesive layer **200** to the display region of the display module **000** reaches the first row of pixels close to the non-display region **01***b*, such that the light emitted from the back light module **300** to the pixels is normally exited from the corresponding pixels.

In the embodiments of the present disclosure, as the barrier wall **301***c* in the frame body **301** can reduce the amount of overflowed adhesive of the adhesive layer **200** in bonding the display panel **100** and the frame body **301**, the width of the overflowed adhesive portion **203** is less, and the maximum thickness of the overflowed adhesive portion **203** is less. Illustratively, in the direction perpendicular to the plane of the light exiting face of the display panel **100**, a maximum thickness h0 of the overflowed adhesive portion **203** in the adhesive layer **200** is less than or equal to twice a minimum distance b between the barrier wall **301***c* and the display panel **100**. Illustratively, in the direction perpendicular to the plane of the light exiting face of the display panel **100**, the maximum thickness h0 of the overflowed adhesive portion **203** in the adhesive layer **200** is less than or equal to 1.6 times of the minimum distance b between the barrier wall **301***c* and the display panel **100**. As such, the refraction of the light by the overflowed adhesive portion **203** is poor, such that the light emitted from the backlight module **300** to the first row of pixels close to the non-display region **01***b* is also normally exited from the row of pixels.

In some embodiments, a shear resistance strength P of the adhesive layer **200** meets:

$$P \geq \frac{m \times g}{12 \times s};$$

m represents a mass of the display panel **100**, g represents gravitational acceleration, and s represents an area of a cross-section of the adhesive layer **200**. It should be noted that the constant **12** in the above formula is a coefficient related to a material of the adhesive layer **200**.

In the present disclosure, in the case that the shear resistance strength P of the adhesive layer **200** meets the above condition, the adhesive layer **200** is not prone to the poor phenomenon of breaking, such that the display panel **100** and the backlight module **300** are fixedly bonded by the adhesive layer **200**.

In some embodiments, in the case that the mass of the display panel **100** is 2.5 kg, and the area of the cross-section of the adhesive layer **200** is 0.4 mm2, the shear resistance strength P of the adhesive layer **200** is greater than or equal to 5.1 MPa.

In the embodiments of the present disclosure, an area S1' of the cross-section of the overflowed adhesive portion **203** in the adhesive layer **200** and an area S2 of a cross-section of the barrier wall **301***c* meet:

$$\frac{\pi \times \left(\frac{H}{0.625}\right)^2}{4} - S1' \leq S2';$$

H represents a minimum distance between the support stage **301***b* and the display panel **100**.

It should be noted that in the case that the barrier wall **301***c* is not disposed in the frame body **301**, the area S0 of the cross-section of the overflowed adhesive portion in the adhesive layer is about:

$$S0 = \frac{\pi \times \left(\frac{H}{0.625}\right)^2}{4};$$

Thus, in the case that the barrier wall **301***c* is disposed in the frame body **301**, a difference value between the area S0 of the cross-section of the overflowed adhesive portion without the barrier wall and the area S1' of the cross-section of the overflowed adhesive portion with the barrier wall is less than or equal to the area S2 of the cross-section of the barrier wall **301***c*.

In the present disclosure, for convenience of calculation of the above area S1' of the cross-section of the overflowed adhesive portion, the shape of the overflowed adhesive portion **203** requires to be analyzed. Referring to FIG. **7**, FIG. **7** is a partially enlarged diagram of another display module according to some embodiments of the present disclosure. The cross-section of the overflowed adhesive portion **203** in the adhesive layer **200** includes an adhesive face S01 and an arc-shaped face S02. The adhesive face S01 is bonded to the display panel **100**, the arc-shaped face S02 is disposed on a side, facing away from the display panel **100**, of the adhesive face S01, and a maximum distance between the arc-shaped face S02 and the display panel **100** is less than or equal to the maximum distance between the support stage **301***b* and the display panel **100**. The maximum distance between the arc-shaped face S02 and the display panel **100** is a maximum thickness h0 of the overflowed adhesive portion **203** in the adhesive layer **200** in the direction perpendicular to the plane of the light exiting face of the display panel **100**.

In some embodiments, the shape of the cross-section of the overflowed adhesive portion **203** in the adhesive layer **200** is approximately composed by a fan **203***a* and a trapezoid **203***b*. The arc-shaped face S02 is an arc-shaped face of the fan **203***a*, and the adhesive face S01 is a face, close to the display panel **100**, of the fan **203***a* and a face, close to the display panel **100**, of the trapezoid **203***b*. A fan-shaped radius angle of the fan **203***a* is about 90°, and a radius r of the fan **203***a* is equal to a maximum thickness of the overflowed adhesive portion **203**. Thus, the radius r of the fan **203***a* is related to the minimum distance b between the barrier wall **301***c* and the display panel **100**. In some embodiments, the radius r of the fan **203***a* is less than or equal to 1.6 times of the minimum distance b between the barrier wall **301***c* and the display panel **100**. A length of an upper base of the trapezoid **203***b* is approximately equal to the distance b between the barrier wall **301***c* and the display panel **100**, and a length of a lower base of the trapezoid **203***b* is approximately equal to the radius r of the fan **203***a*, and a height of the trapezoid **203**b is related to the width of the valid sub-pixel **100**c and the radius r of the fan **203**a. In some embodiments, as the maximum width d0 of the overflowed adhesive portion **203** in the adhesive layer **200** requires to be less than or equal to the overall width of two valid sub-pixels **100**c closest to the edge of the display region **100**a, the height of the trapezoid **203**b is approximately equal to a difference value between the overall width a of two valid sub-pixels **100**c closest to the edge of the display region **100**a and the radius r of the fan **203**a. Thus, a maximum valid area S1 of the cross-section of the overflowed adhesive portion **203** is approximately calculated based on the above relationships. The maximum valid area S1 of the cross-section of the overflowed adhesive portion **203** is an equivalent area of the overflowed adhesive portion **203** in the case that the maximum width of the cross-section of the overflowed adhesive portion **203** is equal to the overall width of two valid sub-pixels **100**c closest to the edge of the display region **100**a.

Thus, the barrier wall **301**c meets the following conditions:

$$\begin{cases} S1 = \dfrac{\pi r^2}{4} + \dfrac{1}{2} \times (b + r) \times (a - r) \\ r = h0 \\ S2 = d_1 \times h \\ \dfrac{\pi \times \left(\dfrac{H}{0.625}\right)^2}{4} - S1 \leqslant S2 \end{cases};$$

H represents a minimum distance between the support stage **301**b and the display panel **100**, d_1 represents a minimum width of the cross-section of the barrier wall **301**c, h represents a minimum height of the barrier wall **301**c in a direction perpendicular to a plane of a light exiting face of the display panel **100**, r represents a radius of the arc-shaped face of the cross-section of the overflowed adhesive portion **203**, h0 represents a maximum thickness of the overflowed adhesive portion **203**, b represents a minimum distance between the barrier wall **301**c and the display panel **100**, a represents the overall width of two valid sub-pixels closest to the edge of the display region **100**a, and S2 represents a valid area of the cross-section of the barrier wall **301**c.

As such, a size of the barrier wall **301**c is predetermined prior to bonding the display panel **100** and the frame body **301** by the adhesive layer **200**, the area of the cross-section of the overflowed adhesive portion **203** in the adhesive layer **200** is predicted upon bonding of the display panel **100** and the support stage **301**b by the adhesive layer **200**, and thus whether the size of the overflowed adhesive portion **203** meets the design requirements is determined.

It should be noted that the embodiments of the present disclosure are illustrated by taking the shape of the cross-section of the barrier wall **301**c being the rectangle as an example. In some embodiments, the shape of the cross-section of the barrier wall **301**c is a regular shape, such as a trapezoid or semicircle, or other irregular shapes, which is not limited in the embodiments of the present disclosure as long as the area of the cross-section of the barrier wall **301**c meets the above conditions.

In some embodiments, the barrier wall **301**c meets the following conditions:

$$\begin{cases} S1 = \dfrac{\pi r^2}{4} + \dfrac{1}{2} \times (b + r) \times (a' - r) \\ r = h0 \\ S2 = d_1 \times h \\ \dfrac{\pi \times \left(\dfrac{H}{0.625}\right)^2}{4} - S1 \leqslant S2 \end{cases};$$

H represents a minimum distance between the support stage **301**b and the display panel **100**, d_1 represents a minimum width of a cross-section of the barrier wall **301**c, h represents a minimum height of the barrier wall **301**c in a direction perpendicular to a plane of a light exiting face of the display panel **100**, r represents a radius of the arc-shaped face of the cross-section of the overflowed adhesive portion **203**, h0 represents a maximum thickness of the overflowed adhesive portion **203**, b represents a minimum distance between the barrier wall **301**c and the display panel **100**, a' represents a distance between an orthogonal projection of an edge, away from the edge of the display region **100**a, of a second valid sub-pixel closest to the edge of the display region **100**a of the display panel **100** on the display panel and an orthogonal projection of the barrier wall **301**c on the display panel **100**, and S2 represents a valid area of the cross-section of the barrier wall **301**c.

It should be noted that the embodiments of the present disclosure are illustrated by taking one barrier wall **301**c being disposed on the support stage **301**b as an example. In some embodiments, there may be a plurality of barrier walls **301**c, which is not limited in the embodiments of the present disclosure as long as a sum of the areas of the cross-section of the barrier walls **301**c on the support stage **301**b meets the above conditions.

Illustratively, referring to FIG. **8**, FIG. **8** is a partially enlarged diagram of another display module according to some embodiments of the present disclosure. At least two barrier walls **301**c are juxtaposed on a face, facing away from the bearing face S, of the support stage **301**b in the frame body **301**, and the barrier walls **301**c are parallel. A gap is defined between any two adjacent barrier walls **301**c, and the adhesive layer further includes an auxiliary adhesive portion **204** in the gap. As such, the affixation in bonding the display panel **100** and the frame body **301** is further improved by the auxiliary adhesive portion **204** in the gap in the adhesive layer **200**.

It should be noted that in the case that at least two barrier walls **301**c are juxtaposed on a face, facing away from the bearing face S, of the support stage **301**b in the frame body **301**, in the direction perpendicular to the plane of the light exiting face of the display panel **100**, the minimum height of the barrier wall **301**c is equal to a minimum height h of the barrier wall **301**c in the above embodiments, and a sum of widths of the cross-sections of the barrier walls **301**c and the minimum width d_1 of the cross-section of the barrier wall **301**c in the above embodiments.

In some embodiments, as shown in FIG. **6**, the side face of the display panel **100** is flush with a face of the edge, close to the display module **000**, of the frame body **301**, or is protruded from the frame body **301**. In some embodiments, an orthogonal projection of the support stage **301**b on the display panel **100** is within the non-display region **100**b of the display panel. In some embodiments, a width d_3 of a side, close to the barrier wall **301**c, of the cross-section of the support stage **301***b* is less than or equal to a width d4 of the non-display region **100***b*. In some embodiments, in the case that a face, away from the edge of the display module **100**, of the support stage **301***b* is a plane, a boundary of the display region **100***a* is flush with the face, away from the edge of the display module **100**, of the support stage **301***b*. As such, the frame body **301** does not block the display region **100***a* of the display panel **100**, such that the display panel **100** normally displays the screen, and the width of the frame of the display module **000** is ensured to be less. It should be noted that being flush of two planes is that the two planes are flushed. While an assembling error exists in practical application, and thus the two planes are not flushed. Thus, one plane is shifted ±0.2 mm relative to the other plane.

In the embodiments of the present disclosure, the display module **000** further includes a plurality of barrier walls **301***c* extending along an edge of the display module **000**. The plurality of barrier walls **301***c* are disposed on a plurality of side faces of the display module, and orthogonal projections of the plurality of barrier walls **301***c* on the display panel **100** surround the display region **100***a*.

In some embodiments, the orthogonal projections of the plurality of barrier walls **301***c* on the display panel **100** surround at least two sides of the display region **100***a*. In some embodiments, in the case that two barrier walls **301***c* are defined, the orthogonal projections of the two barrier walls on the display panel **100** surround two opposite sides or two adjacent sides of the display region **100***a*. In the case that three barrier walls **301***c* are defined, the orthogonal projections of the three barrier walls on the display panel **100** surround three sides of the display region **100***a*. In the case that four barrier walls **301***c* are defined, the orthogonal projections of the four barrier walls on the display panel **100** surround four sides of the display region **100***a*, and in this case, the orthogonal projections of the four barrier walls **301***c* on the display panel **100** surround the display region **100***a*.

Thus, the form of the barrier walls **301***c* in the frame body **301** is achieved in many implementations, and the embodiments of the present disclosure are illustrated in the two possible implementations.

In a first possible implementation, as shown in FIG. **9**, FIG. **9** is a schematic structural diagram of another display module according to some embodiments of the present disclosure. The display module **000** includes a plurality of barrier walls **301***c* extending along an edge of the display module **000**. The plurality of barrier walls **301***c* are disposed on sides of the display module **000**, that is, orthogonal projections of the plurality of barrier walls **301***c* on the display panel **100** surround the display region of the display panel **100**. In some embodiments, the barrier wall **301***c* in the frame body **301** is annular, and the orthogonal projection of the barrier wall **301***c* surrounds the display region **100***a* of the display panel **100**. As such, the barrier wall **301***c* is disposed at any position of the frame body **301**, such that the width of the overflowed adhesive portion **203** at any position of the adhesive layer **200** is less.

In a second possible implementation, as shown in FIG. **10**, FIG. **10** is a schematic structural diagram of another display module according to some embodiments of the present disclosure. The non-display region **100***b* of the display panel **100** includes a bonding region **100***d*. In some embodiments, the display panel **100** includes an array substrate **101** and a color film substrate **102** that are opposite, at least one side of the array substrate **101** protrudes from the color film substrate **102**, and a portion, protruding from the color film substrate **102**, of the array substrate **101** is a portion of the bonding region **100***d*. In the present disclosure, the bonding region **100***d* is disposed on a first side of the display region **100***a*, and the orthogonal projections of the plurality of barrier walls **301***c* in the frame body **301** on the display panel **100** are on other sides of the display region **100***a* than the first side.

In this case, as a width of the side, provided with the bonding region **100***d*, of the non-display region **100***b* of the display panel **100** is great, the amount of the overflowed adhesive of the adhesive layer **200** on the side is ensured to be less even if the side is not provided with the barrier wall **301***c*, and thus the overflowed adhesive portion on the side does not affect the normal display of the display panel **100**. In addition, as widths of other sides, not disposed with the bonding region **100***d*, of the non-display region **100***b* of the display panel **100** are less, the barrier wall **301***c* requires to be disposed on the other sides, and the less amount of the overflowed adhesive of the adhesive layer **200** on the other sides ensures the overflowed adhesive portion on the other sides does not affect the normal display of the display panel **100**.

In the embodiments of the present disclosure, as shown in FIG. **9** and FIG. **10**, the backlight module **300** in the display module **000** further includes: a backplane **303**. The backplane **303** is disposed on a side, facing away from the display panel **100**, of the optical film **302**, the backplane **303** is fixedly connected to the frame body **301**, and the backlight source **304** is disposed between the backplane **303** and the optical film **302**. The backlight source **304** is disposed on the backplane **303**, and is also referred to as a light plate.

In some embodiments, the frame body **301** further includes a connecting portion **301***d* fixedly connected to a side, facing away from the support stage **301***b*, of the bearing stage **301***a*. The backplane **303** includes a backplane body **303***a* configured to support the backplane of the backlight source **304** and a connection portion **303***b* fixedly connected to an edge of the backplane body **303***a*. The connecting portion **301***d* in the frame body **301** is fixedly connected to the connection portion **303***b* in the backplane **303** by a screw F, such that the frame body is fixedly connected to the backplane **303**.

In the present disclosure, the backlight source **304** includes a circuit board **304***a* on a side, close to the optical film **302**, of the backplane body **303***a* and a plurality of light-emitting elements **304***b* that are arranged in an array, disposed on a side, facing away from the backplane body **303***a*, of the circuit board **304***a*, and electrically connected to the circuit board **304***a*. The light-emitting element **304***b* is a light-emitting diode (LED).

In some embodiments, the frame body **301** in the backlight module **300** further includes a surrounding structure **301***e* fixedly connected to the bearing stage **301***a*. The surrounding structure **301***e* surrounds an edge of the backlight source **304**, and a side, close to the backlight source **304**, of the surrounding structure **301***e* is provided with a reflective layer **301***f*. An end, close to the backplane body **303***a*, of the surrounding structure **301***e* limits the edge portion of the circuit board **304***a* in the backlight source **304**.

In the present disclosure, light emitted from the backlight source **304** to the surrounding structure **301***e* is reflected by the reflective layer **301***f*, such that the light is irradiated to the display panel **100** upon passing through the optical film **302**.

In the embodiments of the present disclosure, the backlight module **300** further includes a light guide structure **305** on the bearing face S of the bearing stage **301***a*. The support stage **301***b* in the frame body **301** is closer to the edge of the display module than the light guide structure **305** is, and the optical film **302** in the backlight module **300** is disposed on a side, facing away from the bearing face S, of the light guide structure **305**. The light guide structure **305** is annular. In this case, light emitted from the backlight source **304** to the edge of the display module **000** is guided by the light guide structure **305**, such that the light guided by the light guide structure **305** passes through the optical film **302** and is emitted to pixels arranged in the edge of the display panel **100**, and thus light is normally exited from the edge of the display region **100***a* of the display panel **100**.

In some embodiments, the optical film **302** in the backlight module **300** includes at least one of a diffuse plate **302***a*, a lower prism sheet **302***b*, and an upper prism sheet **302***c*. In some embodiments, the diffuse plate **302***a*, the lower prism sheet **302***b*, and the upper prism sheet **302***c* are laminated in a direction away from the bearing face S.

In the present disclosure, the display panel **100** in the display module **000** further includes a first polarizer **103** on a side, facing away from the color film substrate **102**, of the array substrate **101** and a second polarizer **104** on a side, facing away from the array substrate **101**, of the color film substrate **102**. A polarization direction of the first polarizer **103** is perpendicular to a polarization direction of the second polarizer **104**.

In some embodiments, the display module **000** further includes a chip on flex (COF) **400** bonded with the array substrate **101** and a printed circuit board (PCB) **500** electrically connected to a side, facing away from the array substrate **101**, of the COF **400**. A buffer layer **600** is disposed between the PCB **500** and the connection portion **303***b* in the backplane **303**, and the PCB **500** is buffered by the buffer layer **600** to avoid damage of PCB **400** upon a force on the display module **000**.

In the embodiments of the present disclosure, the display module **000** further includes an auxiliary adhesive layer **700** wrapped in an edge region of the display panel **100** and an edge region of the backlight module **300**. The affixation in assembling the display panel **100** and the backlight module **300** is improved by the auxiliary adhesive layer **700**.

In summary, the display module in the embodiments of the present disclosure includes a display panel, an adhesive layer, and a backlight module. In a frame body of the backlight module, a side, facing away form a bearing face of a bearing stage, of a support stage is disposed with a barrier wall. In bonding the display panel and the frame body by the adhesive layer, an amount of overflowed adhesive from the adhesive layer to a display region of the display module is reduced by the barrier wall, such that a maximum width of a cross-section of an overflowed adhesive portion subsequently formed in the adhesive layer is less. For example, the maximum width of the cross-section of the overflowed adhesive portion in the adhesive layer is less than or equal to an overall width of two valid sub-pixels closest to an edge of the display region of the display panel. Thus, light irradiated to each pixel in light emitted from the backlight module is normally emitted from corresponding pixel. As such, a possibility of poor phenomenon of dark lines in an edge portion of a screen displayed by the display panel is efficiently reduced, and a display effect of the display panel is efficiently improved.

The embodiments of the present disclosure further provide another display module. As shown in FIG. **11**, FIG. **11** is a schematic structural diagram of another display module according to some embodiments of the present disclosure. The display module **000** includes a display panel **100**, an optical film **302**, a frame body **301**, and an adhesive layer **200**.

The optical film **302** in the display module **000** is attached to a light entering face (that is, a face opposite to a display face) of the display panel **100**. The display panel **100** is bonded to the optical film **302** by an optical adhesive layer **800**. For example, the optical film **302** homogenize the backlight.

The frame body **301** in the display module **000** is disposed on a side, facing away from the display panel **100**, of the optical film. The frame body **301** includes a support stage **301***b* and a barrier wall **301***c* on a face, close to the optical film **302**, of the support stage **301***b*.

An outer side face of the support stage **301***b* is protruded from an outer side face of the barrier wall **301***c*, and a width of the barrier wall **301***c* is less than a width of the support stage **301***b*. It should be noted that the support stage **301***b* in the frame body **301** in the display module **000** is generally annular, and thus the outer side face of the support stage **301***b* is a face, facing away from the annular region enclosed by the support stage **301***b*, of the support stage **301***b*, and the outer side face of the barrier wall **301***c* and the outer side face of the support stage **301***b* are disposed on the same side. It should be further noted that the frame body **301** and the optical film **302** in the display module **000** are configured to form the backlight module **300** in the display module **000**.

The adhesive layer **200** in the display module **000** includes a first adhesive portion **201** and a second adhesive portion **202**. The first adhesive portion **201** is disposed between the barrier wall **301***c* in the frame body **301** and the display panel **100** and is bonded to the barrier wall **301***c* and the display panel **100**, and the second adhesive portion **202** is disposed between the support stage **301***b* in the frame body **301** and the display panel **100** and is bonded to the support stage **301***b* and the display panel **100**. The adhesive layer **200** further includes an overflowed adhesive portion **203** on a side, facing away from the second adhesive portion **202**, of the first adhesive portion **201**, and the overflowed adhesive portion **203** in the adhesive layer **200** is bonded to the display panel **100**. The second adhesive portion **202** is closer to the edge of the display module **000** than the first adhesive portion **201** is.

In some embodiments, a minimum width d_1 of a cross-section of the barrier wall and a width d_3 of a side, close to the barrier wall, of a cross-section of the support stage meet: $d_1 \leq d_3/2$.

In some embodiments, a face, close to the optical film, of the barrier wall is flush with a face, close to the optical film, of the support stage.

In some embodiments, the minimum width d_1 of the cross-section of the barrier wall and the width d_3 of the side, close to the barrier wall, of the cross-section of the support stage meet: $d_1 \geq d_3/4$.

In some embodiments, in a direction perpendicular to a plane of a light exiting face of the display panel, a minimum height h of the barrier wall and a minimum distance H between the support stage and the display panel meet: $H/4 \leq h \leq 3H/4$.

In some embodiments, in a direction perpendicular to a plane of a light exiting face of the display panel, a minimum distance between the display panel and the support stage is greater than or equal to 0.2 mm; and/or, in the direction perpendicular to the plane of the light exiting face of the display panel, the minimum distance between the display panel and the support stage is less than or equal to 0.5 mm.

In some embodiments, a maximum width of a cross-section of the adhesive layer is greater than or equal to 1 mm.

In some embodiments, the adhesive layer further includes: an overflowed adhesive portion on a side, facing away from the second adhesive portion, of the first adhesive portion, wherein the overflowed adhesive portion is bonded to the display panel; and

the display panel includes a plurality of valid sub-pixels arranged in an array, and a maximum width of a cross-section of the overflowed adhesive portion is less than or equal to an overall width of two valid sub-pixels closest to an edge of a display region of the display panel.

In some embodiments, in a direction perpendicular to a plane of a light exiting face of the display panel, a maximum thickness of the overflowed adhesive portion is less than or equal to twice a minimum distance between the display panel and the barrier wall.

In some embodiments, the cross-section of the overflowed adhesive portion includes: an adhesive face and an arc-shaped face, wherein the adhesive face is bonded to the display panel, the arc-shaped face is disposed on a side, facing away from the display panel, of the adhesive face, and a maximum distance between the arc-shaped face and the display panel is less than or equal to a maximum distance between the support stage and the display panel.

In some embodiments, an area S1' of the cross-section of the overflowed adhesive portion and an area S2 of a cross-section of the barrier wall meet:

$$\frac{\pi\times\left(\frac{H}{0.625}\right)^2}{4}-S1'\leqslant S2';$$

H represents a minimum distance between the support stage and the display panel.

In some embodiments, relevant dimensions of the barrier wall meet:

$$\begin{cases} S1=\frac{\pi r^2}{4}+\frac{1}{2}\times(b+r)\times(a-r)\\ r=h0\\ S2=d_1\times h\\ \frac{\pi\times\left(\frac{H}{0.625}\right)^2}{4}-S1\leqslant S2 \end{cases};$$

H represents a minimum distance between the support stage and the display panel, d_1 represents a minimum width of a cross-section of the barrier wall, h represents a minimum height of the barrier wall in a direction perpendicular to a plane of a light exiting face of the display panel, r represents a radius of the arc-shaped face, h0 represents a maximum thickness of the overflowed adhesive portion, b represents a minimum distance between the barrier wall and the display panel, and a represents the overall width of two valid sub-pixels closest to the edge of the display region.

In some embodiments, relevant dimensions of the barrier wall meet:

$$\begin{cases} S1=\frac{\pi r^2}{4}+\frac{1}{2}\times(b+r)\times(a'-r)\\ r=h0\\ S2=d_1\times h\\ \frac{\pi\times\left(\frac{H}{0.625}\right)^2}{4}-S1\leqslant S2 \end{cases};$$

H represents a minimum distance between the support stage and the display panel, d_1 represents a minimum width of a cross-section of the barrier wall, h represents a minimum height of the barrier wall in a direction perpendicular to a plane of a light exiting face of the display panel, r represents a radius of the arc-shaped face, h0 represents a maximum thickness of the overflowed adhesive portion, b represents a minimum distance between the barrier wall and the display panel, and a' represents a distance between an orthogonal projection of an edge, away from the edge of the display region, of a second valid sub-pixel closest to the edge of the display region of the display panel on the display panel and an orthogonal projection of the barrier wall on the display panel.

In some embodiments, a shear resistance strength P of the adhesive layer meets:

$$P\geqslant\frac{m\times g}{12\times s};$$

m represents a mass of the display panel, g represents gravitational acceleration, and s represents a minimum area of a cross-section of the adhesive layer.

In some embodiments, the adhesive layer is formed by curing an optical adhesive.

In some embodiments, at least two barrier walls are juxtaposed on a face, facing away from the bearing face, of the support stage, a gap is defined between any two adjacent barrier walls, and the adhesive layer further includes an auxiliary adhesive portion in the gap.

In some embodiments, the display panel further includes a display region and a non-display region on a periphery of the display region, and a width of a side, close to the barrier wall, of a cross-section of the support stage is less than or equal to a width of the non-display region.

In some embodiments, a boundary of the display region is flush with a face, close to the optical film, of the support stage.

In some embodiments, the display module further includes: a plurality of barrier walls extending along an edge of the display module, wherein the plurality of barrier walls are disposed on a plurality of side faces of the display module, and orthogonal projections of the plurality of barrier walls on the display panel surround the display region of the display panel.

In some embodiments, the non-display region includes a bonding region, wherein the bonding region is disposed on a first side of the display region, and the orthogonal projections of the plurality of barrier walls on the display panel are on other sides of the display region than the first side.

In some embodiments, the backlight module further includes: a light guide structure on the bearing face of the bearing stage, wherein the support stage is closer to the edge of the display module than the light guide structure is, and the optical film is disposed on a side, facing away from the bearing face, of the light guide structure.

In some embodiments, the frame body further includes: a surrounding structure fixedly connected to the bearing stage, wherein the surrounding structure surrounds the backlight source, and a side, close to a light plate, of the surrounding structure is provided with a reflective layer.

In some embodiments, the optical film includes: at least one of a diffuse plate, a lower prism sheet, and an upper prism sheet.

It should be noted that the display module in the above embodiments and the display module in the above-mentioned embodiments differ in only a part of the structure, and principles of a less width of the overflowed adhesive portion in the display module are the same. Thus, the principle of the display module is not repeated, which can be referred to the display module in the above-mentioned embodiments.

The embodiments of the present disclosure further provide a backlight module. As shown in FIG. **12**, FIG. **12** is a schematic structural diagram of a backlight module according to some embodiments of the present disclosure. The backlight module **300** includes a backlight source **304**, a frame body **301**, and an optical film **302**.

The frame body **301** in the backlight module **300** includes a bearing stage **301***a*, a support stage **301***b*, and a barrier wall **301***c*. The bearing stage **301***a* in the frame body **301** includes a bearing face S for bearing the optical film **302**. The support stage **301***b* in the frame body **301** is affixed to the bearing face S, and the barrier wall **301***c* in the frame body **301** is affixed to a face, facing away from the bearing face S, of the support stage **301***b*.

It should be noted that the structures and principles of the display module herein can be referred to the related description of the display module in the above-mentioned embodiments, which are not repeated herein.

The embodiments of the present disclosure further provide a display device. The display device includes a plurality of spliced display modules. The display module is the display module shown in FIG. **3**, FIG. **6**, FIG. **7**, FIG. **8**, FIG. **9**, FIG. **10**, or FIG. **11**.

It should be noted that in the accompanying drawings, for clarity of the illustration, the dimension of the layers and regions may be scaled up. It should be understood that when an element or layer is described as being "on" another element or layer, the described element or layer may be directly located on other elements or layers, or an intermediate layer may exist. In addition, it should be understood that when an element or layer is described as being "under" another element or layer, the described element or layer may be directly located under other elements, or more than one intermediate layer or element may exist. In addition, it should be further understood that when a layer or element is described as being arranged "between" two layers or elements, the described layer or element may be the only layer between the two layers or elements, or more than one intermediate layer or element may exist. In the overall disclosure, like reference numerals indicate like elements.

In the present discourse, the terms "first," "second," "third" and the like are only used for the purpose of description and should not be construed as indicating or implying relative importance. Unless otherwise clearly defined, the expression "a plurality of" refers to two or more.

Described above are merely exemplary embodiments of the present disclosure, and are not intended to limit the present disclosure. Any modifications, equivalent replacements, improvements and the like made within the spirit and principles of the present disclosure should be encompassed within the scope of protection of the present disclosure.

The invention claimed is:

1. A display module, comprising:

a display panel;

an adhesive layer; and

a backlight module comprising a backlight source, a frame body, and an optical film, wherein the frame body comprises a bearing stage, a support stage, and a barrier wall, wherein the bearing stage comprises a bearing face configured to bear the optical film, the support stage is affixed to the bearing face, the barrier wall is affixed to a face, facing away from the bearing face, of the support stage, and the barrier wall is disposed on a side, away from an edge of the display module, of the support stage;

wherein the adhesive layer comprises a first adhesive portion, a second adhesive portion, and an overflowed adhesive portion, wherein the first adhesive portion is disposed between the barrier wall and the display panel and is bonded to the barrier wall and the display panel, the second adhesive portion is disposed between the support stage and the display panel and is bonded to the support stage and the display panel, the overflowed adhesive portion is disposed on a side, facing away from the second adhesive portion, of the first adhesive portion and is bonded to the display panel, and the second adhesive portion is closer to an edge of the display module than the first adhesive portion is; and

the display panel comprises a plurality of valid sub-pixels arranged in an array, wherein a maximum width of a cross-section of the overflowed adhesive portion is less than or equal to an overall width of two valid sub-pixels closest to an edge of a display region of the display panel, and the maximum width of the cross-section of the overflowed adhesive portion is greater than 0.

2. The display module according to claim **1**, wherein a minimum width d_1 of a cross-section of the barrier wall and a width d_3 of a side, close to the barrier wall, of a cross-section of the support stage meet: $d_1 \leq d_3/2$.

3. The display module according to claim **2**, wherein a face, close to the optical film, of the barrier wall is flush with a face, close to the optical film, of the support stage.

4. The display module according to claim **3**, wherein the minimum width d_1 of the cross-section of the barrier wall and the width d_3 of the side, close to the barrier wall, of the cross-section of the support stage meet: $d_1 \geq d_3/4$.

5. The display module according to claim **1**, wherein in a direction perpendicular to a plane of a light exiting face of the display panel, a minimum height h of the barrier wall and a minimum distance H between the support stage and the display panel meet: $H/4 \leq h \leq 3H/4$.

6. The display module according to claim **1**, wherein the display module meets at least one of the following requirements:

in a direction perpendicular to a plane of a light exiting face of the display panel, a minimum distance between the display panel and the support stage is greater than or equal to 0.2 mm; and

in the direction perpendicular to the plane of the light exiting face of the display panel, the minimum distance between the display panel and the support stage is less than or equal to 0.5 mm.

7. The display module according to claim **1**, wherein a maximum width of a cross-section of the adhesive layer is greater than or equal to 1 mm.

8. The display module according to claim **1**, wherein in a direction perpendicular to a plane of a light exiting face of the display panel, a maximum thickness of the overflowed adhesive portion is less than or equal to twice a minimum distance between the display panel and the barrier wall.

9. The display module according to claim **1**, wherein the cross-section of the overflowed adhesive portion comprises an adhesive face and an arc-shaped face, wherein the adhesive face is bonded to the display panel, the arc-shaped face is disposed on a side, facing away from the display panel, of the adhesive face, and a maximum distance between the arc-shaped face and the display panel is less than or equal to a maximum distance between the support stage and the display panel.

10. The display module according to claim **9**, wherein an area S1' of the cross-section of the overflowed adhesive portion and an area S2' of a cross-section of the barrier wall meet:

$$\frac{\pi\times\left(\frac{H}{0.625}\right)^2}{4}-S1'\leq S2';$$

wherein H represents a minimum distance between the support stage and the display panel.

11. The display module according to claim **9**, wherein relevant dimensions of the barrier wall meet:

$$\begin{cases} S1=\frac{\pi r^2}{4}+\frac{1}{2}\times(b+r)\times(a-r) \\ r=h0 \\ S2=d_1\times h \\ \frac{\pi\times\left(\frac{H}{0.625}\right)^2}{4}-S1\leq S2 \end{cases};$$

wherein H represents a minimum distance between the support stage and the display panel, d_1 represents a minimum width of a cross-section of the barrier wall, h represents a minimum height of the barrier wall in a direction perpendicular to a plane of a light exiting face of the display panel, r represents a radius of the arc-shaped face, h0 represents a maximum thickness of the overflowed adhesive portion, b represents a minimum distance between the barrier wall and the display panel, and a represents the overall width of two valid sub-pixels closest to the edge of the display region.

12. The display module according to claim **9**, wherein relevant dimensions of the barrier wall meet:

$$\begin{cases} S1=\frac{\pi r^2}{4}+\frac{1}{2}\times(b+r)\times(a'-r) \\ r=h0 \\ S2=d_1\times h \\ \frac{\pi\times\left(\frac{H}{0.625}\right)^2}{4}-S1\leq S2 \end{cases};$$

wherein H represents a minimum distance between the support stage and the display panel, d_1 represents a minimum width of a cross-section of the barrier wall, h represents a minimum height of the barrier wall in a direction perpendicular to a plane of a light exiting face of the display panel, r represents a radius of the arc-shaped face, h0 represents a maximum thickness of the overflowed adhesive portion, b is a minimum distance between the barrier wall and the display panel, and a' represents a distance between an orthogonal projection of an edge, away from the edge of the display region, of a second valid sub-pixel closest to the edge of the display region of the display panel on the display panel and an orthogonal projection of the barrier wall on the display panel.

13. The display panel according to claim **1**, wherein a shear resistance strength P of the adhesive layer meets:

$$P\geq\frac{m\times g}{12\times s};$$

wherein m represents a mass of the display panel, g represents gravitational acceleration, and s represents a minimum area of a cross-section of the adhesive layer.

14. The display module according to claim **1**, wherein at least two barrier walls are juxtaposed on a face, facing away from the bearing face, of the support stage, a gap is defined between any two adjacent barrier walls, and the adhesive layer further comprises an auxiliary adhesive portion in the gap.

15. The display module according to claim **1**, wherein the display panel further comprises a display region and a non-display region on a periphery of the display region, and a width of a side, close to the barrier wall, of a cross-section of the support stage is less than or equal to a width of the non-display region.

16. The display module according to claim **15**, further comprising: a plurality of barrier walls extending along an edge of the display module, wherein the plurality of barrier walls are disposed on a plurality of side faces of the display module, and orthogonal projections of the plurality of barrier walls on the display panel surround the display region of the display panel.

17. The display module according to claim **1**, wherein the backlight module further comprises a light guide structure on the bearing face of the bearing stage, wherein the support stage is closer to the edge of the display module than the light guide structure is, and the optical film is disposed on a side, facing away from the bearing face, of the light guide structure.

18. A display module, comprising:

a display panel;

an optical film attached to a light entering face of the display panel;

a frame body on a side, facing away from the display panel, of the optical film, wherein the frame body comprises a support stage and a barrier wall, wherein the barrier wall is affixed to a face, facing away from the bearing face, of the support stage, and the barrier wall is disposed on a side, away from an edge of the display module, of the support stage; and

an adhesive layer comprising a first adhesive portion, a second adhesive portion, and an overflowed adhesive portion, wherein the first adhesive portion is disposed between the barrier wall and the display panel and is bonded to the barrier wall and the optical film, the second adhesive portion is disposed between the support stage and the display panel and is bonded to the support stage and the optical film, the overflowed adhesive portion is disposed on a side, facing away from the second adhesive portion, of the first adhesive portion and is bonded to the display panel, and the second adhesive portion is closer to an edge of the display module than the first adhesive portion;

wherein the display panel comprises a plurality of valid sub-pixels arranged in an array, wherein a maximum width of a cross-section of the overflowed adhesive portion is less than or equal to an overall width of two valid sub-pixels closest to an edge of a display region of the display panel, and the maximum width of the cross-section of the overflowed adhesive portion is greater than 0.

19. A display device, comprising: a plurality of spliced display modules, wherein the display module comprises:

a display panel;

an adhesive layer; and

a backlight module comprising a backlight source, a frame body, and an optical film, wherein the frame body comprises a bearing stage, a support stage, and a barrier wall, wherein the bearing stage comprises a bearing face configured to bear the optical film, the support stage is affixed to the bearing face, the barrier wall is affixed to a face, facing away from the bearing face, of the support stage, and the barrier wall is disposed on a side, away from an edge of the display module, of the support stage;

wherein the adhesive layer comprises a first adhesive portion, a second adhesive portion, and an overflowed adhesive portion, wherein the first adhesive portion is disposed between the barrier wall and the display panel and is bonded to the barrier wall and the display panel, the second adhesive portion is disposed between the support stage and the display panel and is bonded to the support stage and the display panel, the overflowed adhesive portion is disposed on a side, facing away from the second adhesive portion, of the first adhesive portion and is bonded to the display panel, and the second adhesive portion is closer to an edge of the display module than the first adhesive portion is; and

the display panel comprises a plurality of valid sub-pixels arranged in an array, wherein a maximum width of a cross-section of the overflowed adhesive portion is less than or equal to an overall width of two valid sub-pixels closest to an edge of a display region of the display panel, and the maximum width of the cross-section of the overflowed adhesive portion is greater than 0.

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